

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – ANALYTICAL PROBLEM SOLVING

Course Code:	CSA0615	Course Title:	Design and Analysis of Algorithms
CO Assessed:	CO2 – Manipulation (BL1, BL2, BL3)	Assessment:	Assessment Tool 1 – Analytical Problem Solving
Weightage:	40% of CIA	Total Marks:	100
CO2 Statement:	<i>Apply Brute Force technique to solve sorting, searching, Closest-Pair and Convex-Hull problems (BL1, BL2, BL3)</i>		
Student Name:	V.LASYA	Reg. No.:	192524186
Section / Batch:	BTECH.AI&DS	Date:	31-07-2026

Instructions to Students

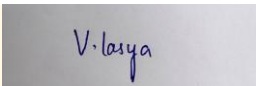
- Answer ALL questions. Total Marks: 100.
- Analyze each problem carefully before applying the appropriate Brute Force technique.
- Clearly show all intermediate steps, comparisons, iterations, and logical reasoning used in solving the problems.
- Apply suitable algorithms for sorting, searching, Closest-Pair, and Convex-Hull problems wherever required.
- Justify the correctness and efficiency of the proposed solution using appropriate complexity analysis.
- Use diagrams, tables, or stepwise illustrations wherever necessary for better explanation.
- Maintain neat presentation, proper algorithmic notation, and structured analytical reasoning throughout the answers.

Question Type	Focus Area	Questions	Marks
Application-Based Questions	Apply Brute Force techniques for sorting, searching, Closest-Pair and Convex-Hull problem analysis	Q1 Q2,Q3	40 60 (2X30)
GRAND TOTAL:		100 Marks	

SECTION A – APPLICATION BASED QUESTIONS**Code of Conduct**

I ____ V.Lasya(192524186) _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student:



Q38. Bubble Sort – Alternating High and Low Values

Apply the Brute Force technique using Bubble Sort on the dataset [50, 10, 40, 20, 30, 5]. Clearly interpret the problem and explain how Bubble Sort repeatedly exchanges adjacent out-of-order elements. Perform the sorting process pass by pass and show all intermediate arrays. Count the total number of comparisons and swaps made during execution. Analyze the time complexity in best, average, and worst cases. Verify the final sorted output and interpret the efficiency of the algorithm on alternating values.

Bubble Sort - Alternating High & Low Values

1. problem Interpretation

- Bubble sort is a simple Brute Force sorting algorithm
- It works by comparing adjacent elements in the array
- If two adjacent elements are in the wrong order, they are swapped
- After every pass, the largest unsorted element moves (bubbles) to its correct position at the end of the array
- The process continues until no more swaps are required and the array becomes completely sorted.

Given Dataset

Input Array: [50, 10, 40, 20, 30, 5]

Pass - by - Pass sorting

Initial Array

• [50, 10, 40, 20, 30, 5]

Pass 1

[10, 40, 20, 30, 5, 50]

Pass 2 : [10, 20, 30, 5, 40, 50]
Pass 3 : [10, 20, 5, 30, 40, 50]
Pass 4 : [10, 5, 20, 30, 40, 50]
Pass 5 : [5, 10, 20, 30, 40, 50]

Intermediate Arrays

Initial : [50, 10, 40, 20, 30, 5]
After pass 1 : [10, 40, 20, 30, 5, 50]
After pass 2 : [10, 20, 30, 5, 40, 50]
After pass 3 : [10, 20, 5, 30, 40, 50]
After pass 4 : [10, 5, 20, 30, 40, 50]
After pass 5 : [5, 10, 20, 30, 40, 50]

Comparisons and Swaps

pass 1 : 5 comparisons, 5 swaps
pass 2 : 4 comparisons, 3 swaps
pass 3 : 3 comparisons, 1 swap
pass 4 : 2 comparisons, 1 swap
pass 5 : 1 comparison, 1 swap

Total comparisons = 15

Total swaps = 11

Time Complexity

Best case : $O(n)$ - When the array is already sorted and an optimized Bubble sort stops after one pass

Average case : $O(n^2)$ - Random order of Elements

Worst case : $O(n^2)$ - Reverse sorted Array

Space complexity : $O(1)$ - Because it sorts the array in place without using extra memory

Final sorted Output

Sorted Array : [5, 10, 20, 30, 40, 50]

Verification

- The output is in ascending order
- Every element is correctly placed according to its value
- Therefore, the Bubble sort algorithm has successfully sorted the given dataset.

Efficiency Analysis

- Bubble sort is easy to understand and implement
- It is suitable for small datasets
- It performs many unnecessary comparisons for large arrays, making it inefficient
- Since it only swaps adjacent elements, an element may require several passes to reach its correct position.

→ For large datasets, algorithm such as Merge sort, Quick sort or Heap sort are preferred because they have an average time complexity of $O(n \log n)$.

Interpretation on Alternating Values

→ The given array contains alternating large and small values

→ Such as arrangement creates many adjacent inversions

→ As a result, Bubble sort performs a high no. of swaps (11) before sorting is complete.

→ This demonstrates that Bubble sort is less efficient for alternating or highly unsorted data because elements move only one position at a time.

RUBRICS FOR CALCULATION

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Problem Context	20%	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30%	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	20%	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20%	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10%	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand

Marks Evaluation:

Question No.	Understanding of Problem Context (20%)	Application of Concepts (30%)	Problem Solving Approach (20%)	Accuracy of Solution (20%)	Clarity (10%)	Total Marks
Q1 (40 Marks)	4	4	4	4	4	40
Q2 (30 Marks)	4	4	4	4	4	30
Q3 (30 Marks)	4	3	3	2	2	22

Overall Total (100)	92
----------------------------	-----------

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____